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Audience	Core Working Group (CWG), previously Evolution Working Group (EWG)

Permitting trivial default initialization in constexpr contexts

Updates over Revision 1

- Updated wording based on feedback from Jens

Updates over Revision 0

- Revision 0 passed the Evolution Working Group straw poll and was forwarded to Core Working Group for inclusion in C++2a

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- Merged content from "Open questions" section into "Avoiding undefined behavior" section and removed the "Open questions" section as there are no more open questions
- Added wording
- Minor edits

Introduction

This paper proposes permitting default initialization for trivially default constructible types in constexpr contexts while continuing to disallow the invocation of undefined behavior. In short, so long as uninitialized values are not read from, such states should be permitted in constexpr in both heap and stack allocated scenarios.

Prerequisite

Let the below `NontrivialType` and `TrivialType` type definitions be imported and visible for all following example code snippets.

```
struct NontrivialType {  
    bool val = false;  
};
```

```
struct TrivialType {  
    bool val;  
};
```

Justification as of C++17

For types with nontrivial default constructors, default initialization is defined as calling the default constructor. This behavior is consistent across runtime and constexpr contexts (assuming the default constructor is marked as or defaulted as constexpr). Unfortunately, this guarantee does not generically apply across the language. For types with trivial default constructors, default initialization does not compile in constexpr. This becomes a pain point when writing generic code. Some constexpr function templates that are well defined for types with user-defined constexpr default constructors will fail to compile when instantiated with, for example, any of the fundamental integer types provided by the language.

The below code example, `Example1`, shows how this inconsistency can negatively affect code that one would expect to be consistent across template instantiations. If this proposal is adopted, all four below callsites, instead of just the first three, would be valid and compile covering the full matrix of runtime/constexpr by trivially/nontrivially default constructible.

```
namespace Example1 {

template <typename T>
constexpr T f(const T& other) {
    T t;                // default initialization
    t = other;
    return t;
}

// These three successfully compile
auto nontrivial_rt = f(NontrivialType{});
auto trivial_rt = f(TrivialType{});
constexpr auto nontrivial_ct = f(NontrivialType{});

// As of C++17, this fails to compile meaning `Example1::f(const T&)` is not
// generic
constexpr auto trivial_ct = f(TrivialType{});

} // namespace Example1
```

Note: `nontrivial_rt` is constant initialized since `Example1::f(const NontrivialType&)` is valid in constexpr. As a bonus, by making `Example1::f(const TrivialType&)` valid in constexpr, `trivial_rt` will also be given constant initialization.

Justification for C++2a

In addition to the unexpectedly inconsistent behavior demonstrated above, the presence of P0784 [\[1\]](#) in C++2a strengthens the need for this proposed change. Trivial default initialization is allowed, but only for heap allocated objects. This proposal would simply fix the "bug" disallowing that same behavior for stack allocated objects.

The below examples, `Example2` and `Example3`, show side-by-side what should be semantically equivalent functions. The current C++2a draft standard, with the adoption of P0784, would permit, in constexpr, the function definitions that use heap allocation but not the function definitions that use stack allocation. This proposal, if adopted with P0784, would result in all four below callsites being valid.

```
namespace Example2 {

template <typename T>
constexpr T f1(const T& other) {
    auto* t = new T;        // default initialization
    *t = other;
    T out = *t;
    delete t;
    return out;
}

template <typename T>
constexpr T f2(const T& other) {
    T t;                    // default initialization
    t = other;
    T out = t;
    return out;
}
```

```

}

// Successfully compiles
constexpr auto trivial_ct_h = f1(TrivialType{});

// Fails to compile meaning `Example2::f1(const T&)` and
// `Example2::f2(const T&)` are observably different
constexpr auto trivial_ct_s = f2(TrivialType{});

} // namespace Example2

```

```

namespace Example3 {

template <typename T>
constexpr T f1(const T& other) {
    auto* t = new T[1];           // default initialization
    t[0] = other;
    T out = t[0];
    delete[] t;
    return out;
}

template <typename T>
constexpr T f2(const T& other) {
    T t[1];                       // default initialization
    t[0] = other;
    T out = t[0];
    return out;
}

// Successfully compiles
constexpr auto trivial_ct_h = f1(TrivialType{});

// Fails to compile meaning `Example3::f1(const T&)` and
// `Example3::f2(const T&)` are observably different
constexpr auto trivial_ct_s = f2(TrivialType{});

} // namespace Example3

```

Avoiding undefined behavior

As always, undefined behavior cannot be invoked in constexpr. In order to allow trivial default initialization, some mechanism must be in place that prevents reading uninitialized values. That mechanism should continue to be compilation failure; however, instead of the overly restrictive current behavior (failing to compile at the point of initialization), compilation should only fail at the point in code where the uninitialized read takes place.

This should not be difficult for compiler developers to implement nor should it cause much if any compile time overhead. The steps required are similar to detecting reads of out-of-lifetime variables, something that current compilers already implement for constant evaluation. [\[2\]](#)

That said, reading uninitialized instances of `unsigned char` at runtime is not undefined. For now, such behavior shall remain non-constant. From Richard Smith:

The most reasonable thing would be to say that (in the short term at least), reading the value of an uninitialized object is non-constant regardless of its type. We can revisit that decision if we find a use case, but going in the other direction would be more problematic due to being a potentially breaking change. [\[3\]](#)

Further justification: On "Constexpr all the things!"

`absl::InlinedVector<T, N, A>` is a `std::vector<T, A>`-like type that takes advantage of the Small Size Optimization (SSO). SSO is implemented by including a buffer inside the type to avoid allocation for small instances. As an optimization, unused bytes in the inlined buffer are left uninitialized. "Don't pay for what you don't use," as they say.

In the spirit of P0784 [1], if InlinedVector were to be made available in constexpr, the behavior should be consistent with the behavior of runtime instances. The unused inlined bytes should be able to be left uninitialized so that implementers may be able to expose accidental uninitialized reads through constexpr evaluation and UBSan.

Additionally, from a maintenance perspective, not having to fork the implementation to handle constexpr is much appreciated.

Wording

To ensure reading uninitialized instances of `unsigned char` remains non-constant, under **[expr.const]**, add a new bullet to **p4** between the existing **(4.8)** and **(4.9)**:

```
[...]
(4.8) - an lvalue-to-rvalue conversion that is applied to a glvalue that refers to a
       non-active member of a union or a subobject thereof;
+ (4.X) - an lvalue-to-rvalue conversion that is applied to an object with indeterminate
+       value ([basic.indet]);
(4.9) - an invocation of an implicitly-defined copy/move constructor or copy/move assignment
       operator for a union whose active member (if any) is mutable, unless the lifetime of
       the union object began within the evaluation of e;
[...]
```

Under **[dcl.constexpr]**, remove the last condition from **(3.4.4)** and update the example:

```
[...]
(3.4.4) - a definition of a variable of non-literal type or of static or thread storage
-       duration or for which no initialization is performed.
+       duration.
[...]
```

```
[...]
constexpr int uninit() {
-   int a;           // error: variable is uninitialized
-   return a;
+   struct { int a; } s;
+   return s.a;      // error: uninitialized read of s.a
}
[...]
```

Also under **[dcl.constexpr]**, remove **(4.4)**:

```
[...]
(4.3) - either its function-body shall be = default, or the compound-statement of its
       function-body shall satisfy the requirements for a function-body of a constexpr function;
- (4.4) - every non-variant non-static data member and base class subobject shall be initialized
-       ([class.base.init]);
(4.5) - if the class is a union having variant members ([class.union]), exactly one of them shall
       be initialized;
[...]
```

Under **[cpp.predefined]**, update `__cpp_constexpr` in table 17 under **(1.8)**:

Macro name	Value
...	
__cpp_conditional_explicit	201806L
- __cpp_constexpr	201603L
+ __cpp_constexpr	201907L
__cpp_coroutines	201902L
...	

References

[1] [P0784: More constexpr containers](#)

[2] Personal correspondence with Richard Smith about this proposal

[3] [Comment by Richard Smith in \[RFC\] P1331: Permitting trivial default initialization in constexpr contexts](#)